

Question ID: 9a144a01

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

Which of the following is true about the values of 2^x and $2x+2$ for $x > 0$?

Answer

- A. For all $x > 0$, it is true that $2^x < 2x+2$.
- B. For all $x > 0$, it is true that $2^x > 2x+2$.
- C. There is a constant c such that if $0 < x < c$, then $2^x < 2x+2$, but if $x > c$, then $2^x > 2x+2$.
- D. There is a constant c such that if $0 < x < c$, then $2^x > 2x+2$, but if $x > c$, then $2^x < 2x+2$.

Correct Answer: C

Rationale

Choice C is correct. At $x = 0$, the value of 2^x is less than the value of $2x+2$: $2^0 < 2(0)+2$, which is equivalent to $1 < 2$. As the value of x increases, the value of 2^x remains less than the value of $2x+2$ until $x = 3$, which is when the two values are equal: $2^3 = 2(3)+2$, which is equivalent to $8 = 8$. Then, for $x > 3$, the value of 2^x is greater than the value of $2x+2$. So there is a constant, 3, such that when $0 < x < 3$, then $2^x < 2x+2$, but when $x > 3$, then $2^x > 2x+2$.

Choice A is incorrect because $2^x > 2x+2$ when $x > 3$. Choice B is incorrect because $2^x < 2x+2$ when $0 < x < 3$. Choice D is incorrect because $2^x < 2x+2$ when $0 < x < 3$ and $2^x > 2x+2$ when $x > 3$.